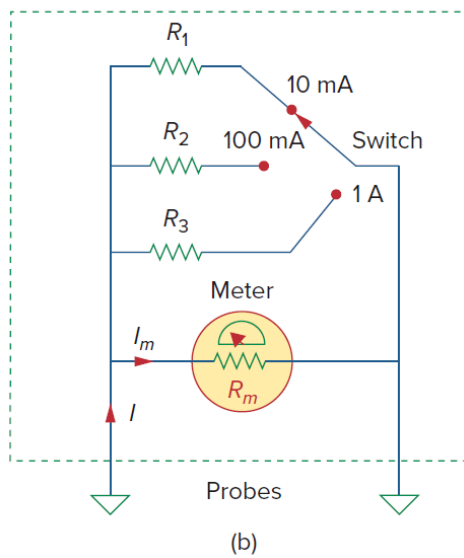
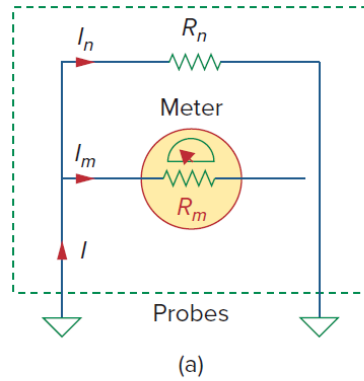


Problem

Following the ammeter setup of Fig. 2.61, design an ammeter for the following multiple ranges: (a) 0–1 A (b) 0–100 mA (c) 0–10 mA Take the full-scale meter current as $I_m = 1$ mA and the internal resistance of the ammeter as $R_m = 50\ \Omega$.

Figure 2.61

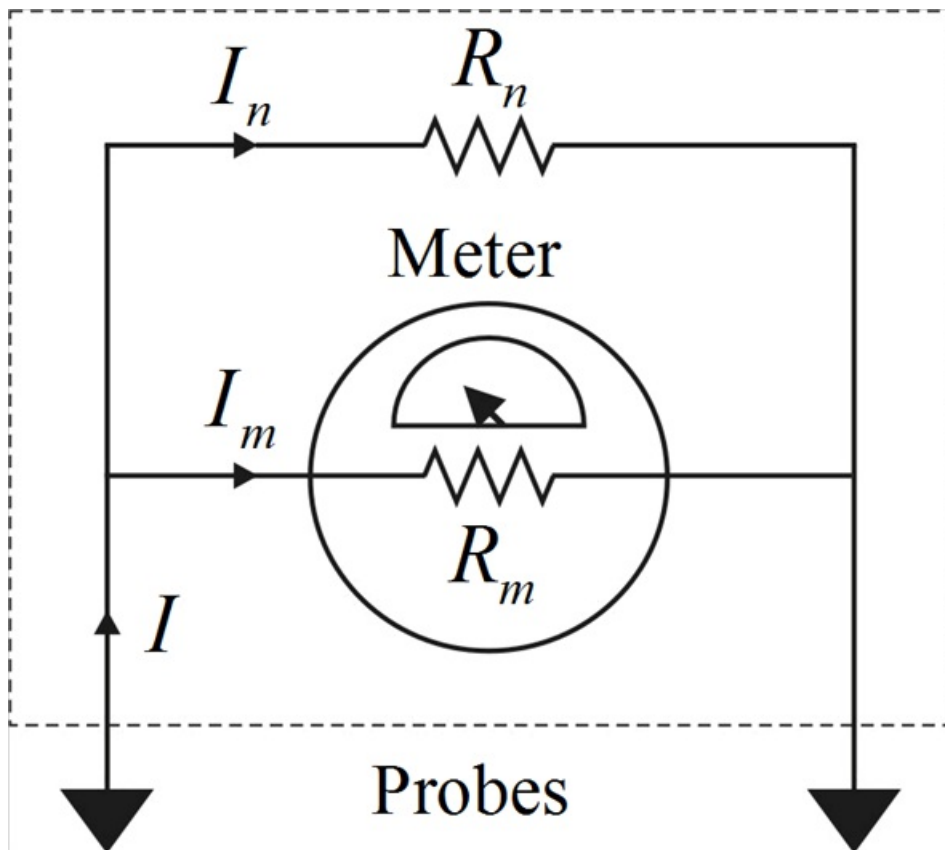
Ammeters: (a) single-range type, (b) multiple-range type.



Step-by-step solution

Step 1 of 5

The given Ammeter setup is:



Step 2 of 5

From the above network, we have

Full scale reading $I = I_{fs} = I_m + I_n$

Applying the current division principle yields,

$$I_m = \frac{R_n}{R_n + R_m} I_{fs}$$

$$R_n = \frac{I_m}{I_{fs} - I_m} R_m \dots\dots\dots (1)$$

Full scale meter current, $I_m = 1 \text{ mA}$

Internal resistance of the ammeter, $R_m = 50 \Omega$

Step 3 of 5

(a) The given range is 0 – 1A

From the given range, full scale reading is, $I_{fs} = 1 \text{ A}$.

Substituting all the values in equation (1), shunt resistance is obtained as:

$$R_n = \frac{1 \times 10^{-3}}{1 - (1 \times 10^{-3})} \times 50$$

$$\therefore R_n = 0.05 \Omega$$

Step 4 of 5

(b) The given range is 0 – 100 mA

From the given range, full scale reading is, $I_{fs} = 100 \text{ mA}$.

Substituting all the values in equation (1), shunt resistance is obtained as:

$$R_n = \frac{1 \times 10^{-3}}{(100 \times 10^{-3}) - (1 \times 10^{-3})} \times 50$$

$$R_n = \frac{1}{99} \times 50$$

$$\therefore R_n = 0.505 \, \Omega$$

Step 5 of 5

(c) The given range is 0 – 10 mA

From the given range, full scale reading is, $I_{fs} = 10 \text{ mA}$.

Substituting all the values in equation (1), shunt resistance is obtained as:

$$R_n = \frac{1 \times 10^{-3}}{(10 \times 10^{-3}) - (1 \times 10^{-3})} \times 50$$

$$R_n = \frac{1}{9} \times 50$$

$$\therefore R_n = 5.556 \, \Omega$$